

To start with, let's look at a 2×2 determinant:

$A = \begin{vmatrix} a & b \\ c & d \end{vmatrix}$. The value of A is $ad - bc$. (see p. 815)

Eg. $\begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} = 1(4) - 3(2) = 4 - 6 = -2$.

- A determinant can be positive, negative or zero.

The cross product of $\vec{a}$ and $\vec{b}$ is a vector perpendicular to both $\vec{a}$ and $\vec{b}$ with direction according to the right-hand rule (see p. 816)

Therefore: $\vec{a} \times \vec{b} \perp \vec{a}$ and $\vec{a} \times \vec{b} \perp \vec{b}$.

Definition (p. 815)

If $\vec{a} = \langle a_1, a_2, a_3 \rangle$ and $\vec{b} = \langle b_1, b_2, b_3 \rangle$, then the cross product (or vector product) of $\vec{a}$ and $\vec{b}$ is the vector:

$$\begin{aligned} \vec{a} \times \vec{b} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} \\ &= \vec{i} \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} - \vec{j} \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} + \vec{k} \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} \\ &= \langle a_2 b_3 - a_3 b_2, -(a_1 b_3 - a_3 b_1), a_1 b_2 - a_2 b_1 \rangle \end{aligned}$$

Theorems: Let $\vec{a}, \vec{b} \in \mathbb{R}^3$ and $\vec{a} \neq \vec{0}, \vec{b} \neq \vec{0}$.

1) Theorem 8 (p. 816)

$\vec{a} \times \vec{b}$ is perpendicular to both $\vec{a}$ and $\vec{b}$.

It is worth-while looking at the proof of this theorem on p. 816.

2) Theorem 9 (p. 817)

$$|\vec{a} \times \vec{b}| = |\vec{a}| |\vec{b}| \sin \theta \quad (\text{see p. 817})$$

3) Corollary 10 (p. 817)

$\vec{a}$ and $\vec{b}$ are parallel $\Leftrightarrow \vec{a} \times \vec{b} = \vec{0}$

($\Leftrightarrow$: if and only if.)

4) $|\vec{a} \times \vec{b}|$ is the area of the parallelogram determined by $\vec{a}$ and $\vec{b}$.

Properties MS (p. 819)

The commutative property does not hold, but the distributive property holds (see properties 3 and 4)

Examples:

1 $\vec{a} = \langle 2, 1, 3 \rangle$, $\vec{b} = \langle 1, 2, 1 \rangle$

(a) $\vec{a} \times \vec{b} = \langle 2, 1, 3 \rangle \times \langle 1, 2, 1 \rangle$

$$= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 1 & 3 \\ 1 & 2 & 1 \end{vmatrix}$$

$$= \vec{i}(1-6) - \vec{j}(2-3) + \vec{k}(4-1)$$

$$= \langle -5, 1, 3 \rangle.$$

Now also determine $\vec{b} \times \vec{a}$, to verify property 1.

(b) Now find: $\vec{a} \times 2\vec{b}$ and use it to determine $2\vec{b} \times \vec{a}$

$$\vec{a} \times 2\vec{b} = 2(\vec{a} \times \vec{b}) = 2\langle -5, 1, 3 \rangle = \langle -10, 2, 6 \rangle$$

$$2\vec{b} \times \vec{a} = -(\vec{a} \times 2\vec{b}) = \langle 10, -2, -6 \rangle$$

(c) Determine $\vec{a} \times \vec{a}$

2. Find a unit vector perpendicular to both $\vec{a} = \langle 2, 0, 0 \rangle$ and $\vec{b} = \langle 0, 3, 0 \rangle$.

$$\vec{a} \times \vec{b} = \langle 2, 0, 0 \rangle \times \langle 0, 3, 0 \rangle$$

$$= 6 \langle 1, 0, 0 \rangle \times \langle 0, 1, 0 \rangle$$

$$= 6 \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{vmatrix} = 6\vec{i}(0-0) - 6\vec{j}(0-0) + 6\vec{k}(1-0)$$

$$= \langle 0, 0, 6 \rangle$$

Unit vector:

$$\frac{\langle 0, 0, 6 \rangle}{|\langle 0, 0, 6 \rangle|} = \frac{\langle 0, 0, 6 \rangle}{\sqrt{0+0+36}} = \frac{1}{6} \langle 0, 0, 6 \rangle = \langle 0, 0, 1 \rangle$$

Another unit vector perpendicular to both $\vec{a}$ and $\vec{b}$ is

$$-\langle 0, 0, 1 \rangle.$$

A vector with length 2, perpendicular to both $\vec{a}$ and $\vec{b}$, is: $2\langle 0, 0, 1 \rangle$ or $-2\langle 0, 0, 1 \rangle$

(3)

MS: Do examples 3 and 4 on p. 818.

You may omit learning outcomes 6 and 7 of Unit 6.3